

WASP-104b

R-2557

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-104_170331 · created 2026-08-26T22:50:36+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2457843.79049 +/- 0.0008 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.148 +/- 0.04
Transit depth (Rp/Rs)^2	2.19 +/- 1.19 [%]
Orbital Inclination (inc)	81.37 +/- 2.79
Airmass coefficient 1 (a1)	0.874 +/- 0.014
Airmass coefficient 2 (a2)	0.116 +/- 0.028
Scatter in the residuals of the lightcurve fit is	1.6 %
Best Comparison Star	#1 - [489, 155]
Optimal Aperture	4.08
Optimal Annulus	7.21
Transit Duration (day)	0.046 +/- 0.034

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.148 ± 0.04 vs 0.1214 (deviation 0.56σ)
Duration fit vs published	0.046 d vs 0.0742 d (deviation 0.7σ)
Mid-time O–C	1.9 min
Depth SNR	3.1
Residual scatter	1.6 %

Light curve

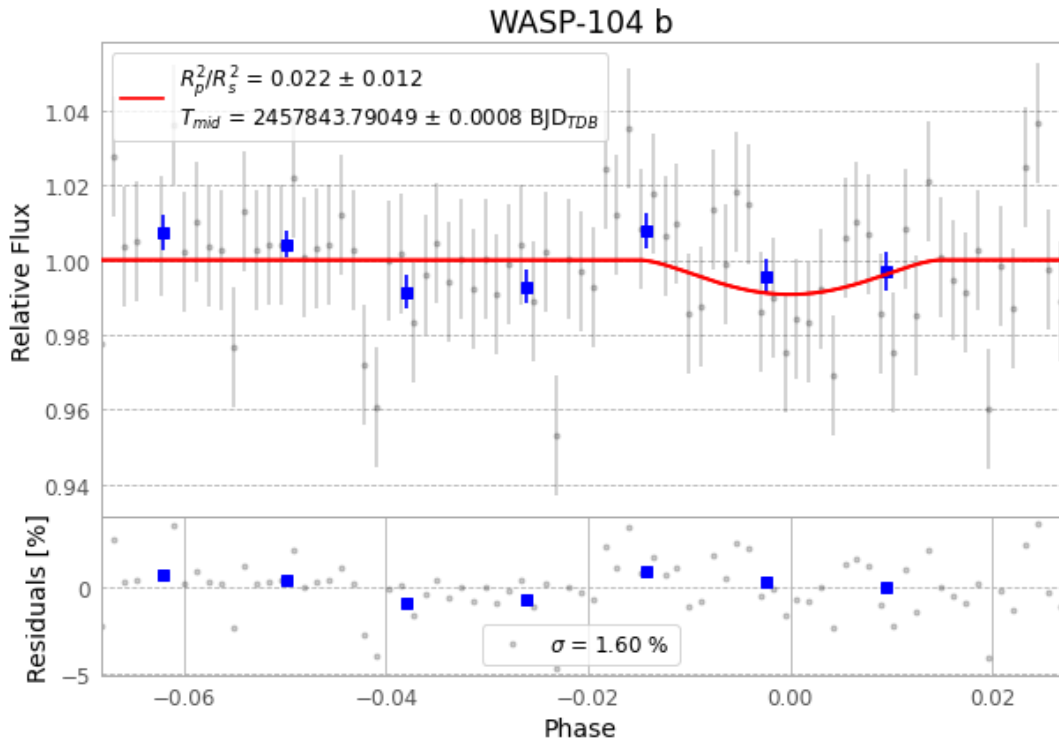


Figure 1 — FinalLightCurve_WASP-104 b_2017-03-30.png (SHA-256 7b32e4205344430a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [104, 256], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 6ac7ed99be0f4958...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

81 FITS frames (53 MB), WASP-104170331035712.FITS → WASP-104170331075712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-104-170331.log (57b7fd28c10a...), FinalLightCurve_WASP-104 b_2017-03-30.png (7b32e4205344...), FinalLightCurve_WASP-104 b_2017-03-30.csv (8385e7e7cca5...)

Compute resources

Wall time 154.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-26 22:50Z.